Drake vs Kendrick Rhyme Scheme Analysis

How does lyrical complexity, measured through rhyme density and lyrical complexity, relate to popularity and listener engagement in hip-hop? Specifically, how do these patterns differ between two of the genre's most influential artists, Kendrick Lamar and Drake, across their careers?

In [1]:
```
import os
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sb
from collections import Counter
import nltk
from nltk.corpus import cmudict
from collections import Counter
nltk.download("cmudict") # dictionary for rhyme scheme analysis
from dotenv import load_dotenv
import lyricsgenius as lg
import requests
import spacy
import re
import time
import sqlite3
nlp=spacy.load("en_core_web_sm")
```

```
[nltk_data] Downloading package cmudict to
[nltk_data]     /Users/mattabatangle/nltk_data...
[nltk_data]   Package cmudict is already up-to-date!
```

In [2]:
```
load_dotenv("3510.env")
```

Out[2]: True

Test to make sure stuff is working

In [3]:
```
geniusClientAccess = os.getenv("GENIUS_CLIENT_ACCESS_TOKEN")
clientID = os.getenv("SPOTIFY_CLIENT_ID")
genius = lg.Genius(geniusClientAccess)
genius.remove_section_headers = True
```

In [4]:
```
temp = genius.search_artist("Kendrick Lamar", max_songs=5) # test
```

```
Searching for songs by Kendrick Lamar...

Song 1: "Not Like Us"
Song 2: "HUMBLE."
Song 3: "euphoria"
Song 4: "m.A.A.d city"
Song 5: "Swimming Pools (Drank)"

Reached user-specified song limit (5).
Done. Found 5 songs.
```

In [5]:
```
genius.search_artist("Drake", max_songs=5) # test
```

```
Searching for songs by Drake...

Song 1: "God’s Plan"
Song 2: "In My Feelings"
Song 3: "Hotline Bling"
Song 4: "One Dance"
Song 5: "Hold On, We’re Going Home"

Reached user-specified song limit (5).
Done. Found 5 songs.
```

Out[5]:
```
Artist(name=Drake, num_songs=5)
```

In [6]:
```
clientID = os.getenv("SPOTIFY_CLIENT_ID")
clientSecret = os.getenv("SPOTIFY_CLIENT_SECRET")
grantType = 'client_credentials'
bodyParams = {"grant_type": grantType}
url = 'https://accounts.spotify.com/api/token'
responseJSON = requests.post(url, data=bodyParams, auth=(clientID, clientSec
headers = {"Authorization": "Bearer "+responseJSON['access_token']}
rootURL = 'https://api.spotify.com/v1/'
```

Gathering each artists unique ID will make things easier when scraping their entire catelogs.

In [7]:
```
kendrickSearch = requests.get(rootURL+"search?q=Kendrick-Lamar&type=artist",
```

In [8]:
```
kendrickID = kendrickSearch['artists']['items'][0]['id']
```

In [9]:
```
drakeSearch = requests.get(rootURL+"search?q=Drake&type=artist",headers=heac
```

In [10]:
```
drakeID = drakeSearch['artists']['items'][0]['id']
```

Now lets get started

First, I will scrape every available song from the Spotify API for both artists. This will make it easy to then pull 30-40 indivudal songs that will give an accurate representation of each artists discography over time.

In [11]:
```
def get_all_albums_and_singles(artist_id, include_groups=("album", "single")
    albums = [] # list to hold
    params = {"limit":50, "include_groups": ",".join(include_groups), "marke
    url = rootURL+f"artists/{artist_id}/albums" # base url
    while url:
        albumSearch = requests.get(url, headers=headers, params=params)
        if albumSearch.status_code != 200: # Troubleshoot
            print(f"Error: {albumSearch.status_code} for {url}")
            break
        data = albumSearch.json()
        albums.extend(data["items"]) # Extend instead of append to add multi
        url = data["next"] # Iterate

        params = None # Params already set as of here
    return albums
```

In [12]:
```
kendrickAlbumsSingles = get_all_albums_and_singles(kendrickID)
drakeAlbumsSingles = get_all_albums_and_singles(drakeID)
```

In [13]:
```
def get_album_tracks(album_id):
    tracks = []
    url = f"{rootURL}albums/{album_id}/tracks" # base url
    params = {"limit": 50, "market": "US"} # reasonable limit of 50 songs pe

    while url:
        trackReq = requests.get(url, headers=headers, params=params)
        if trackReq.status_code != 200: # Troubleshoot
            print(f"Error {trackReq.status_code} fetching tracks for album {
            break
        data = trackReq.json()
        tracks.extend(data["items"]) # Extend instead of append to add multi
        url = data["next"] # Iterate
        params = None # Params already set as of here
    return tracks
```

Next I will iterate through each artists albums and gather track data

In [14]:
```
rows = []
for album in kendrickAlbumsSingles:
    albumID = album["id"]
    albumName = album["name"]
    releaseDate = album["release_date"]
    albumType = album["album_group"] or album["album_type"]

    tracks = get_album_tracks(albumID)

    for t in tracks:
        if not any(a["id"] == kendrickID for a in t["artists"]):
            continue
        rows.append({
            "artist": "Kendrick Lamar",
```

```
            "artist_id": kendrickID,
            "album_id": albumID,
            "album_name": albumName,
            "album_type": albumType,
            "release_date": releaseDate,
            "track_id": t["id"],
            "track_name": t["name"],
            "duration_ms": t["duration_ms"],
            "track_number": t["track_number"],
            "disc_number": t["disc_number"],
            "explicit": t.get("explicit", False),
        })
```

In [15]:
```
kendrickTracksUnfiltered = pd.DataFrame(rows)
kendrickTracksUnfiltered
```

Out[15]:

	artist	artist_id	album_id	album_name	alb
0	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	0hvT3yIEysuuvkK73vgdcW	GNX	
1	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	0hvT3yIEysuuvkK73vgdcW	GNX	
2	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	0hvT3yIEysuuvkK73vgdcW	GNX	
3	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	0hvT3yIEysuuvkK73vgdcW	GNX	
4	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	0hvT3yIEysuuvkK73vgdcW	GNX	
...	...	...	...	...	
166	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	3qHsYfN7ua1WGhQn1wquIn	Push Thru (feat. Kendrick Lamar and Curren$y)	
167	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	3qHsYfN7ua1WGhQn1wquIn	Push Thru (feat. Kendrick Lamar and Curren$y)	
168	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	3qHsYfN7ua1WGhQn1wquIn	Push Thru (feat. Kendrick Lamar and Curren$y)	
169	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	1Jmy19ydnxfHcOdN28cw7q	My People (feat. Jay Rock) ("Bastards of the P...	
170	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	3X2elJMfPKM7pXo855g1dy	She Needs Me [Remix] (Ringtone)	

171 rows × 12 columns

In [16]:

```
rows = []
for album in drakeAlbumsSingles:
    albumID = album["id"]
    albumName = album["name"]
    releaseDate = album["release_date"]
    albumType = album["album_group"] or album["album_type"]

    tracks = get_album_tracks(albumID)
```

```
    for t in tracks:
        # Skip tracks where artist isn't primary
        if not any(a["id"] == drakeID for a in t["artists"]):
            continue
        rows.append({
            "artist": "Drake",
            "artist_id": drakeID,
            "album_id": albumID,
            "album_name": albumName,
            "album_type": albumType,
            "release_date": releaseDate,
            "track_id": t["id"],
            "track_name": t["name"],
            "duration_ms": t["duration_ms"],
            "track_number": t["track_number"],
            "disc_number": t["disc_number"],
            "explicit": t.get("explicit", False) # default false
        })
```

In [17]:
```
drakeTracksUnfiltered = pd.DataFrame(rows)
drakeTracksUnfiltered
```

Out[17]:

	artist	artist_id	album_id	album_name	album
0	Drake	3TVXtAsR1Inumwj472S9r4	6Rl6YoCarF2GHPSQmmFjuR	*ome*xy $ongs 4 U	
1	Drake	3TVXtAsR1Inumwj472S9r4	6Rl6YoCarF2GHPSQmmFjuR	*ome*xy $ongs 4 U	
2	Drake	3TVXtAsR1Inumwj472S9r4	6Rl6YoCarF2GHPSQmmFjuR	*ome*xy $ongs 4 U	
3	Drake	3TVXtAsR1Inumwj472S9r4	6Rl6YoCarF2GHPSQmmFjuR	*ome*xy $ongs 4 U	
4	Drake	3TVXtAsR1Inumwj472S9r4	6Rl6YoCarF2GHPSQmmFjuR	*ome*xy $ongs 4 U	
...	...	...	...	...	
397	Drake	3TVXtAsR1Inumwj472S9r4	4VVBDjcpEc8vFWMlQfnmHX	Miss Me	
398	Drake	3TVXtAsR1Inumwj472S9r4	645FRpimwu0b1isiY5M32J	I'm Goin In	
399	Drake	3TVXtAsR1Inumwj472S9r4	2wmvVAqB2sAiJQpp2pRWcD	Fear	
400	Drake	3TVXtAsR1Inumwj472S9r4	4Z4vthIzd6pIKC396R5vSN	Forever	
401	Drake	3TVXtAsR1Inumwj472S9r4	3aMtvuGUNyIR27bv8gFPXw	Best I Ever Had	

402 rows × 12 columns

Next I want to select songs from each artist (top 20/30 + 10 low-pop)

There is a small issue though. Popularity score is not avaliable when being called through a /albums/{id}/tracks endpoint. Therefore, we need to access each track through a /tracks/{id} endpoint to get its popularity score and add it to the rows.

In [18]:
```
def add_popularity(df):
    ids = df["track_id"].dropna().unique().tolist() # Get a unique list of a
    pops = {}
    for i in range(0, len(ids), 50):  # Spotify lets up to 50 per call
        chunk = ids[i:i+50]  # Spotify API only allows up to 50 IDs per requ
        popRequest = requests.get(rootURL + "tracks", headers=headers, param
        if popRequest.status_code == 200:
            for track in popRequest.json()["tracks"]:
                if track:
                    pops[track["id"]] = track.get("popularity", None) # Acco
    df["popularity"] = df["track_id"].map(pops) # Map the popularity dictior
    return df
```

In [19]:
```
add_popularity(kendrickTracksUnfiltered)
```

Out[19]:

	artist	artist_id	album_id	album_name	alb
0	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	0hvT3yIEysuuvkK73vgdcW	GNX	
1	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	0hvT3yIEysuuvkK73vgdcW	GNX	
2	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	0hvT3yIEysuuvkK73vgdcW	GNX	
3	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	0hvT3yIEysuuvkK73vgdcW	GNX	
4	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	0hvT3yIEysuuvkK73vgdcW	GNX	
...	...	...	...	...	
166	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	3qHsYfN7ua1WGhQn1wquIn	Push Thru (feat. Kendrick Lamar and Curren$y)	
167	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	3qHsYfN7ua1WGhQn1wquIn	Push Thru (feat. Kendrick Lamar and Curren$y)	
168	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	3qHsYfN7ua1WGhQn1wquIn	Push Thru (feat. Kendrick Lamar and Curren$y)	
169	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	1Jmy19ydnxfHcOdN28cw7q	My People (feat. Jay Rock) ("Bastards of the P...	
170	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	3X2elJMfPKM7pXo855g1dy	She Needs Me [Remix] (Ringtone)	

171 rows × 13 columns

In [20]:
```
add_popularity(drakeTracksUnfiltered)
```

Out[20]:

	artist	artist_id	album_id	album_name	album
0	Drake	3TVXtAsR1Inumwj472S9r4	6Rl6YoCarF2GHPSQmmFjuR	*ome*xy $ongs 4 U	
1	Drake	3TVXtAsR1Inumwj472S9r4	6Rl6YoCarF2GHPSQmmFjuR	*ome*xy $ongs 4 U	
2	Drake	3TVXtAsR1Inumwj472S9r4	6Rl6YoCarF2GHPSQmmFjuR	*ome*xy $ongs 4 U	
3	Drake	3TVXtAsR1Inumwj472S9r4	6Rl6YoCarF2GHPSQmmFjuR	*ome*xy $ongs 4 U	
4	Drake	3TVXtAsR1Inumwj472S9r4	6Rl6YoCarF2GHPSQmmFjuR	*ome*xy $ongs 4 U	
...	...	...	...	...	
397	Drake	3TVXtAsR1Inumwj472S9r4	4VVBDjcpEc8vFWMlQfnmHX	Miss Me	
398	Drake	3TVXtAsR1Inumwj472S9r4	645FRpimwu0b1isiY5M32J	I'm Goin In	
399	Drake	3TVXtAsR1Inumwj472S9r4	2wmvVAqB2sAiJQpp2pRWcD	Fear	
400	Drake	3TVXtAsR1Inumwj472S9r4	4Z4vthIzd6pIKC396R5vSN	Forever	
401	Drake	3TVXtAsR1Inumwj472S9r4	3aMtvuGUNyIR27bv8gFPXw	Best I Ever Had	

402 rows × 13 columns

In [21]:

```
def select_tracks_for_analysis(df, top_n, low_n):
    df = df.dropna(subset=["track_id", "popularity"]).drop_duplicates("track
    top = (df.sort_values("popularity", ascending=False).groupby("album_id",
    top = top.sort_values("popularity", ascending=False).head(top_n)

    low = (df.sort_values("popularity", ascending=True).drop_duplicates("tra
    low = (low.groupby("album_id", as_index=False).head(2).sort_values("popu

    selectedTracks = (pd.concat([top, low], ignore_index=True).drop_duplicat
    return selectedTracks
```

In [22]:

```
kendrickTracks = select_tracks_for_analysis(kendrickTracksUnfiltered, top_n=
drakeTracks = select_tracks_for_analysis(drakeTracksUnfiltered, top_n=30, lo
```

In [23]:

```
kendrickTracks
```

Out[23]:

	artist	artist_id	album_id	album_name	al
0	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	3pLdWdkj83EYfDN6H2N8MR	Black Panther The Album Music From And Inspire...	
1	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	5JjnoGJyOxfSZUZtk2rRwZ	Not Like Us	
2	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	4eLPsYPBmXABThSJ821sqY	DAMN.	
3	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	0hvT3yIEysuuvkK73vgdcW	GNX	
4	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	4eLPsYPBmXABThSJ821sqY	DAMN.	
5	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	2XPrwlaAHHXnJzP9tBcIzH	Swimming Pools (Drank)	
6	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	0hvT3yIEysuuvkK73vgdcW	GNX	
7	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	4eLPsYPBmXABThSJ821sqY	DAMN.	
8	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	4eLPsYPBmXABThSJ821sqY	DAMN.	
9	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	6PBZN8cbwkqm1ERj2BGXJ1	good kid, m.A.A.d city	
10	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	0hvT3yIEysuuvkK73vgdcW	GNX	
11	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	7ycBtnsMtyVbbwTfJwRjSP	To Pimp A Butterfly	
12	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	4eLPsYPBmXABThSJ821sqY	DAMN.	
13	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	79ONNoS4M9tfIA1mYLBYVX	Mr. Morale & The Big Steppers	
14	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	0hvT3yIEysuuvkK73vgdcW	GNX	
15	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	79ONNoS4M9tfIA1mYLBYVX	Mr. Morale & The Big Steppers	
16	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	32bR4LcEc1PvJEhaKoo4ZN	euphoria	

	artist	artist_id	album_id	album_name	al
17	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	79ONNoS4M9tfIA1mYLBYVX	Mr. Morale & The Big Steppers	
18	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	7ycBtnsMtyVbbwTfJwRjSP	To Pimp A Butterfly	
19	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	3DGQ1iZ9XKUQxAUWjfC34w	good kid, m.A.A.d city (Deluxe)	
20	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	3X2elJMfPKM7pXo855g1dy	She Needs Me [Remix] (Ringtone)	
21	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	3qHsYfN7ua1WGhQn1wquln	Push Thru (feat. Kendrick Lamar and Curren$y)	
22	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	3qHsYfN7ua1WGhQn1wquln	Push Thru (feat. Kendrick Lamar and Curren$y)	
23	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	4NuKTiNjwcX4ZG7TXEBnYh	Malcolm X	
24	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	4XF4wNh0WHyrayqCYS8xY0	Bitch, Don't Kill My Vibe (Remix)	
25	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	5moHeRF6YwQz9Mhc1SB0v0	Two Presidents	
26	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	0itierEc1U3I8yIi4TtnuT	The Mantra	
27	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	4alcGHjstaALJHHiljfy3H	DAMN. COLLECTORS EDITION.	
28	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	1Jmy19ydnxfHcOdN28cw7q	My People (feat. Jay Rock) ("Bastards of the P...	
29	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	6gyBOHTEtKHEPIy9EAWjdB	Swimming Pools (Drank) [Black Hippy Remix]	

In [24]:
```
drakeTracks
```

Out[24]:

	artist	artist_id	album_id	album_name	a
0	Drake	3TVXtAsR1Inumwj472S9r4	40GMAhriYJRO1rsY4YdrZb	Views	
1	Drake	3TVXtAsR1Inumwj472S9r4	6X1x82kppWZmDzlXXK3y3q	Take Care (Deluxe)	
2	Drake	3TVXtAsR1Inumwj472S9r4	1lXY618HWkwYKJWBRYR4MK	More Life	
3	Drake	3TVXtAsR1Inumwj472S9r4	1ATL5GLyefJaxhQzSPVrLX	Scorpion	
4	Drake	3TVXtAsR1Inumwj472S9r4	6OQ9gBfg5EXeNAEwGSs6jK	Dark Lane Demo Tapes	
5	Drake	3TVXtAsR1Inumwj472S9r4	6Rl6YoCarF2GHPSQmmFjuR	*ome*xy $ongs 4 U	
6	Drake	3TVXtAsR1Inumwj472S9r4	40GMAhriYJRO1rsY4YdrZb	Views	
7	Drake	3TVXtAsR1Inumwj472S9r4	6Rl6YoCarF2GHPSQmmFjuR	*ome*xy $ongs 4 U	
8	Drake	3TVXtAsR1Inumwj472S9r4	3SpBlxme9WbeQdI9kx7KAV	Certified Lover Boy	
9	Drake	3TVXtAsR1Inumwj472S9r4	1lXY618HWkwYKJWBRYR4MK	More Life	
10	Drake	3TVXtAsR1Inumwj472S9r4	40GMAhriYJRO1rsY4YdrZb	Views	
11	Drake	3TVXtAsR1Inumwj472S9r4	2ZUFSbIkmFkGag000RWOpA	Nothing Was The Same (Deluxe)	
12	Drake	3TVXtAsR1Inumwj472S9r4	3cf4iSSKd8ffTncbtKljXw	Honestly, Nevermind	
13	Drake	3TVXtAsR1Inumwj472S9r4	1ATL5GLyefJaxhQzSPVrLX	Scorpion	
14	Drake	3TVXtAsR1Inumwj472S9r4	6Rl6YoCarF2GHPSQmmFjuR	*ome*xy $ongs 4 U	
15	Drake	3TVXtAsR1Inumwj472S9r4	4TWJJZmgYm6tE6PzqpluQ0	Which One	
16	Drake	3TVXtAsR1Inumwj472S9r4	1LShhEEKRT5MNPcO7jtYHh	So Far Gone	
17	Drake	3TVXtAsR1Inumwj472S9r4	7dqpveMVcWgbzqYrOdkFTD	Care Package	
18	Drake	3TVXtAsR1Inumwj472S9r4	3SpBlxme9WbeQdI9kx7KAV	Certified Lover Boy	
19	Drake	3TVXtAsR1Inumwj472S9r4	6OQ9gBfg5EXeNAEwGSs6jK	Dark Lane Demo Tapes	
20	Drake	3TVXtAsR1Inumwj472S9r4	6OQ9gBfg5EXeNAEwGSs6jK	Dark Lane Demo Tapes	
21	Drake	3TVXtAsR1Inumwj472S9r4	0bNfrGST1xSPZ3V3g5jjHT	What Did I Miss?	
22	Drake	3TVXtAsR1Inumwj472S9r4	6X1x82kppWZmDzlXXK3y3q	Take Care (Deluxe)	
23	Drake	3TVXtAsR1Inumwj472S9r4	6X1x82kppWZmDzlXXK3y3q	Take Care (Deluxe)	
24	Drake	3TVXtAsR1Inumwj472S9r4	6X1x82kppWZmDzlXXK3y3q	Take Care (Deluxe)	
25	Drake	3TVXtAsR1Inumwj472S9r4	6X1x82kppWZmDzlXXK3y3q	Take Care (Deluxe)	

	artist	artist_id	album_id	album_name	a
26	Drake	3TVXtAsR1Inumwj472S9r4	4czdORdCWP9umpbhFXK2fW	For All The Dogs	
27	Drake	3TVXtAsR1Inumwj472S9r4	5LuoozUhs2pl3glZeAJl89	Scary Hours 2	
28	Drake	3TVXtAsR1Inumwj472S9r4	40GMAhriYJRO1rsY4YdrZb	Views	
29	Drake	3TVXtAsR1Inumwj472S9r4	1lXY618HWkwYKJWBRYR4MK	More Life	
30	Drake	3TVXtAsR1Inumwj472S9r4	3CHCm9n05a3tG4cQJ4rJwq	For All The Dogs Scary Hours Edition (Instrume...	
31	Drake	3TVXtAsR1Inumwj472S9r4	3CHCm9n05a3tG4cQJ4rJwq	For All The Dogs Scary Hours Edition (Instrume...	
32	Drake	3TVXtAsR1Inumwj472S9r4	0v4UlHEDcUBbV9kM3TX4bj	Live For	
33	Drake	3TVXtAsR1Inumwj472S9r4	4VVBDjcpEc8vFWMlQfnmHX	Miss Me	
34	Drake	3TVXtAsR1Inumwj472S9r4	2lzV5TlBiLED8eCjwsGBlv	MELTDOWN (feat. Drake) [Instrumental]	
35	Drake	3TVXtAsR1Inumwj472S9r4	4OQEGCbts7MiJm8wfwC9BX	LOYAL (feat. Drake)	
36	Drake	3TVXtAsR1Inumwj472S9r4	7iLdPmU6eUTwKsYifBQ6zb	TWIST & TURN (feat. Drake & PARTYNEXTDOOR)	
37	Drake	3TVXtAsR1Inumwj472S9r4	4v2wkI8xYxp0gTtSMwbRoj	She Will	
38	Drake	3TVXtAsR1Inumwj472S9r4	4P9NQ88JRt2r39QlsevM1n	Make Me Proud	
39	Drake	3TVXtAsR1Inumwj472S9r4	4v2wkI8xYxp0gTtSMwbRoj	She Will	

Next I want to retrieve the lyrics for each song for my rhyme scheme anaylsis.

In [25]:
```
def get_lyrics_safe(artist, title):
    try: # In case of failed lyric retrieval
        song = genius.search_song(title, artist)
        if song and song.lyrics:
            return song.lyrics
    except Exception as e: # Error handling
        print(f"Error for {artist} - {title}: {e}")
    return None
```

In [26]:
```
def fetch_lyrics(df):
    df = df.copy() # Copy of df
```

```
    df["lyrics"] = None
    for i, row in df.iterrows(): # Iterate thru rows to scrape tracks for ly
        lyrics = get_lyrics_safe(row["artist"], row["track_name"])
        df.at[i, "lyrics"] = lyrics # at index i set lyrics to lyrics
        print(f"{i+1}/{len(df)}: {row['artist']} - {row['track_name']}") # F
        time.sleep(5)  # In case of rate limits
    return df
```

In [27]:
```
kendrickLyrics = fetch_lyrics(kendrickTracks)
drakeLyrics = fetch_lyrics(drakeTracks)
```

```
Searching for "All The Stars (with SZA) - From "Black Panther: The Album"" b
y Kendrick Lamar...
No results found for: 'All The Stars (with SZA) - From "Black Panther: The A
lbum" Kendrick Lamar'
1/30: Kendrick Lamar - All The Stars (with SZA) - From "Black Panther: The A
lbum"
Searching for "Not Like Us" by Kendrick Lamar...
Done.
2/30: Kendrick Lamar - Not Like Us
Searching for "HUMBLE." by Kendrick Lamar...
Done.
3/30: Kendrick Lamar - HUMBLE.
Searching for "tv off (feat. lefty gunplay)" by Kendrick Lamar...
Done.
4/30: Kendrick Lamar - tv off (feat. lefty gunplay)
Searching for "LOVE. FEAT. ZACARI." by Kendrick Lamar...
Done.
5/30: Kendrick Lamar - LOVE. FEAT. ZACARI.
Searching for "Swimming Pools (Drank)" by Kendrick Lamar...
Done.
6/30: Kendrick Lamar - Swimming Pools (Drank)
Searching for "luther (with sza)" by Kendrick Lamar...
No results found for: 'luther (with sza) Kendrick Lamar'
7/30: Kendrick Lamar - luther (with sza)
Searching for "PRIDE." by Kendrick Lamar...
Done.
8/30: Kendrick Lamar - PRIDE.
Searching for "DNA." by Kendrick Lamar...
Done.
9/30: Kendrick Lamar - DNA.
Searching for "Money Trees" by Kendrick Lamar...
Done.
10/30: Kendrick Lamar - Money Trees
Searching for "squabble up" by Kendrick Lamar...
Done.
11/30: Kendrick Lamar - squabble up
Searching for "Alright" by Kendrick Lamar...
Done.
12/30: Kendrick Lamar - Alright
Searching for "LOYALTY. FEAT. RIHANNA." by Kendrick Lamar...
Done.
13/30: Kendrick Lamar - LOYALTY. FEAT. RIHANNA.
Searching for "Count Me Out" by Kendrick Lamar...
Done.
14/30: Kendrick Lamar - Count Me Out
Searching for "peekaboo (feat. azchike)" by Kendrick Lamar...
Done.
15/30: Kendrick Lamar - peekaboo (feat. azchike)
Searching for "United In Grief" by Kendrick Lamar...
Done.
16/30: Kendrick Lamar - United In Grief
Searching for "euphoria" by Kendrick Lamar...
Done.
17/30: Kendrick Lamar - euphoria
Searching for "Die Hard" by Kendrick Lamar...
Error for Kendrick Lamar - Die Hard: Request timed out:
```

```
HTTPSConnectionPool(host='api.genius.com', port=443): Read timed out. (read
timeout=5)
18/30: Kendrick Lamar - Die Hard
Searching for "King Kunta" by Kendrick Lamar...
Done.
19/30: Kendrick Lamar - King Kunta
Searching for "Poetic Justice" by Kendrick Lamar...
Done.
20/30: Kendrick Lamar - Poetic Justice
Searching for "She Needs Me [Remix] (Ringtone)" by Kendrick Lamar...
No results found for: 'She Needs Me [Remix] (Ringtone) Kendrick Lamar'
21/30: Kendrick Lamar - She Needs Me [Remix] (Ringtone)
Searching for "Push Thru - Instrumental" by Kendrick Lamar...
No results found for: 'Push Thru - Instrumental Kendrick Lamar'
22/30: Kendrick Lamar - Push Thru - Instrumental
Searching for "Push Thru" by Kendrick Lamar...
Done.
23/30: Kendrick Lamar - Push Thru
Searching for "Malcolm X" by Kendrick Lamar...
Done.
24/30: Kendrick Lamar - Malcolm X
Searching for "Bitch, Don’t Kill My Vibe - Remix" by Kendrick Lamar...
Done.
25/30: Kendrick Lamar - Bitch, Don’t Kill My Vibe - Remix
Searching for "Two Presidents" by Kendrick Lamar...
Done.
26/30: Kendrick Lamar - Two Presidents
Searching for "The Mantra - From "Creed II: The Album"" by Kendrick Lamar...
No results found for: 'The Mantra - From "Creed II: The Album" Kendrick Lama
r'
27/30: Kendrick Lamar - The Mantra - From "Creed II: The Album"
Searching for "BLOOD." by Kendrick Lamar...
Done.
28/30: Kendrick Lamar - BLOOD.
Searching for "My People (feat. Jay Rock) ("Bastards of the Party" Original
Soundtrack Version)" by Kendrick Lamar...
No results found for: 'My People (feat. Jay Rock) ("Bastards of the Party" O
riginal Soundtrack Version) Kendrick Lamar'
29/30: Kendrick Lamar - My People (feat. Jay Rock) ("Bastards of the Party"
Original Soundtrack Version)
Searching for "Swimming Pools (Drank) - Black Hippy Remix" by Kendrick Lama
r...
No results found for: 'Swimming Pools (Drank) - Black Hippy Remix Kendrick L
amar'
30/30: Kendrick Lamar - Swimming Pools (Drank) - Black Hippy Remix
Searching for "One Dance" by Drake...
Done.
1/40: Drake - One Dance
Searching for "Headlines" by Drake...
Done.
2/40: Drake - Headlines
Searching for "Passionfruit" by Drake...
Done.
3/40: Drake - Passionfruit
Searching for "God's Plan" by Drake...
Done.
```

```
4/40: Drake - God's Plan
Searching for "Not You Too (feat. Chris Brown)" by Drake...
Done.
5/40: Drake - Not You Too (feat. Chris Brown)
Searching for "NOKIA" by Drake...
Done.
6/40: Drake - NOKIA
Searching for "Hotline Bling" by Drake...
Done.
7/40: Drake - Hotline Bling
Searching for "DIE TRYING" by Drake...
Done.
8/40: Drake - DIE TRYING
Searching for "Fair Trade (with Travis Scott)" by Drake...
No results found for: 'Fair Trade (with Travis Scott) Drake'
9/40: Drake - Fair Trade (with Travis Scott)
Searching for "Teenage Fever" by Drake...
Done.
10/40: Drake - Teenage Fever
Searching for "9" by Drake...
Done.
11/40: Drake - 9
Searching for "Hold On, We're Going Home" by Drake...
Done.
12/40: Drake - Hold On, We're Going Home
Searching for "Jimmy Cooks (feat. 21 Savage)" by Drake...
Done.
13/40: Drake - Jimmy Cooks (feat. 21 Savage)
Searching for "Nice For What" by Drake...
Done.
14/40: Drake - Nice For What
Searching for "SOMEBODY LOVES ME" by Drake...
Done.
15/40: Drake - SOMEBODY LOVES ME
Searching for "Which One (feat. Central Cee)" by Drake...
Done.
16/40: Drake - Which One (feat. Central Cee)
Searching for "Best I Ever Had" by Drake...
Done.
17/40: Drake - Best I Ever Had
Searching for "Trust Issues" by Drake...
Done.
18/40: Drake - Trust Issues
Searching for "Yebba’s Heartbreak" by Drake...
Done.
19/40: Drake - Yebba’s Heartbreak
Searching for "Time Flies" by Drake...
Done.
20/40: Drake - Time Flies
Searching for "Chicago Freestyle (feat. Giveon)" by Drake...
Done.
21/40: Drake - Chicago Freestyle (feat. Giveon)
Searching for "What Did I Miss?" by Drake...
Done.
22/40: Drake - What Did I Miss?
Searching for "Take Care" by Drake...
```

```
Done.
23/40: Drake - Take Care
Searching for "Marvins Room" by Drake...
Done.
24/40: Drake - Marvins Room
Searching for "Practice" by Drake...
Done.
25/40: Drake - Practice
Searching for "Over My Dead Body" by Drake...
Done.
26/40: Drake - Over My Dead Body
Searching for "Rich Baby Daddy (feat. Sexyy Red & SZA)" by Drake...
Done.
27/40: Drake - Rich Baby Daddy (feat. Sexyy Red & SZA)
Searching for "Wants and Needs (feat. Lil Baby)" by Drake...
Done.
28/40: Drake - Wants and Needs (feat. Lil Baby)
Searching for "Feel No Ways" by Drake...
Done.
29/40: Drake - Feel No Ways
Searching for "Do Not Disturb" by Drake...
Done.
30/40: Drake - Do Not Disturb
Searching for "Evil Ways (feat. J. Cole) - Instrumental" by Drake...
Specified song does not contain lyrics. Rejecting.
31/40: Drake - Evil Ways (feat. J. Cole) - Instrumental
Searching for "The Shoe Fits - Instrumental" by Drake...
Specified song does not contain lyrics. Rejecting.
32/40: Drake - The Shoe Fits - Instrumental
Searching for "Live For" by Drake...
Done.
33/40: Drake - Live For
Searching for "Miss Me" by Drake...
Done.
34/40: Drake - Miss Me
Searching for "MELTDOWN (feat. Drake) - Instrumental" by Drake...
Specified song does not contain lyrics. Rejecting.
35/40: Drake - MELTDOWN (feat. Drake) - Instrumental
Searching for "LOYAL (feat. Drake)" by Drake...
Done.
36/40: Drake - LOYAL (feat. Drake)
Searching for "TWIST & TURN (feat. Drake & PARTYNEXTDOOR)" by Drake...
Done.
37/40: Drake - TWIST & TURN (feat. Drake & PARTYNEXTDOOR)
Searching for "She Will" by Drake...
Done.
38/40: Drake - She Will
Searching for "Make Me Proud" by Drake...
Done.
39/40: Drake - Make Me Proud
Searching for "She Will - Instrumental" by Drake...
Specified song does not contain lyrics. Rejecting.
40/40: Drake - She Will - Instrumental
```

Genius was not able to find lyrics for 5-10 songs for each artists so lets remove those

In [28]:
```
kendrickLyrics = kendrickLyrics[kendrickLyrics["lyrics"].notna()].reset_inde
drakeLyrics = drakeLyrics[drakeLyrics["lyrics"].notna()].reset_index(drop=Tr

print(len(kendrickLyrics), "Kendrick tracks with lyrics")
print(len(drakeLyrics), "Drake tracks with lyrics")
```

```
22 Kendrick tracks with lyrics
35 Drake tracks with lyrics
```

Next Step is Rhyme & Complexity Analysis

In [29]:
```
prondict = cmudict.dict()
```

In [52]:
```
# CMU dictionary for phonetic rhyme analysis
```

In [31]:
```
# Return a tuple of phonemes for the rhyming part of a word
def rhyme_part(word):
    word = word.lower() # Set standard to lower
    if word in prondict:
        phones = prondict[word][0] # Match word in dictionary
        vowels = [i for i, p in enumerate(phones) if p[-1].isdigit()] # grab
        # Creates a list of indices where the phoneme ends with a digit, whi
        print(vowels)
        return tuple(phones[vowels[-1]:]) if vowels else tuple(phones[-2:])
        # If the word has vowel phonemes, return the phonemes from the last
    return tuple(word[-2:])  # Fallback if not found then use last 2 letters
    # tuple because most hashable for later use (better for dictionary usage
```

In [32]:
```
rhyme_part('aaberg')
```

```
[0, 2]
```

Out[32]:
```
('ER0', 'G')
```

In [33]:
```
def rhyme_density(lyrics):
    words = re.findall(r"\b[a-zA-Z']+\b", str(lyrics).lower()) # regex to cl
    if len(words) < 10:
        return 0
    rhymes = [rhyme_part(w) for w in words] # Set rhymes using last functior
    rhyme_counts = Counter(rhymes) # Count rhymes
    repeating = sum(v for v in rhyme_counts.values() if v > 1) # Find repeat
    return repeating / len(words) # Return density
```

In [63]:
```
# Add word count, rhyme density, and overall complexity stats to df of songs
def add_rhyme_stats(df):
    df = df.copy() # copy df
    df["word_count"] = df["lyrics"].str.split().apply(len) # Count the total
    df["rhyme_density"] = df["lyrics"].apply(rhyme_density) # Count rhyme de
```

```
    df["complexity"] = df["rhyme_density"] * df["word_count"] # Complexity =
    return df
```

In [ ]:
```
kendrickStats = add_rhyme_stats(kendrickLyrics)
drakeStats = add_rhyme_stats(drakeLyrics)
```

In [ ]:
```
kendrickStats
```

In [37]:
```
drakeStats
```

Out[37]:

	artist	artist_id	album_id	album_name	a
0	Drake	3TVXtAsR1Inumwj472S9r4	40GMAhriYJRO1rsY4YdrZb	Views	
1	Drake	3TVXtAsR1Inumwj472S9r4	6X1x82kppWZmDzlXXK3y3q	Take Care (Deluxe)	
2	Drake	3TVXtAsR1Inumwj472S9r4	1lXY618HWkwYKJWBRYR4MK	More Life	
3	Drake	3TVXtAsR1Inumwj472S9r4	1ATL5GLyefJaxhQzSPVrLX	Scorpion	
4	Drake	3TVXtAsR1Inumwj472S9r4	6OQ9gBfg5EXeNAEwGSs6jK	Dark Lane Demo Tapes	
5	Drake	3TVXtAsR1Inumwj472S9r4	6Rl6YoCarF2GHPSQmmFjuR	omeexy $ongs 4 U	
6	Drake	3TVXtAsR1Inumwj472S9r4	40GMAhriYJRO1rsY4YdrZb	Views	
7	Drake	3TVXtAsR1Inumwj472S9r4	6Rl6YoCarF2GHPSQmmFjuR	omeexy $ongs 4 U	
8	Drake	3TVXtAsR1Inumwj472S9r4	1lXY618HWkwYKJWBRYR4MK	More Life	
9	Drake	3TVXtAsR1Inumwj472S9r4	40GMAhriYJRO1rsY4YdrZb	Views	
10	Drake	3TVXtAsR1Inumwj472S9r4	2ZUFSbIkmFkGag000RWOpA	Nothing Was The Same (Deluxe)	
11	Drake	3TVXtAsR1Inumwj472S9r4	3cf4iSSKd8ffTncbtKljXw	Honestly, Nevermind	
12	Drake	3TVXtAsR1Inumwj472S9r4	1ATL5GLyefJaxhQzSPVrLX	Scorpion	
13	Drake	3TVXtAsR1Inumwj472S9r4	6Rl6YoCarF2GHPSQmmFjuR	omeexy $ongs 4 U	
14	Drake	3TVXtAsR1Inumwj472S9r4	4TWJJZmgYm6tE6PzqpluQ0	Which One	

	artist	artist_id	album_id	album_name	a
15	Drake	3TVXtAsR1Inumwj472S9r4	1LShhEEKRT5MNPcO7jtYHh	So Far Gone	
16	Drake	3TVXtAsR1Inumwj472S9r4	7dqpveMVcWgbzqYrOdkFTD	Care Package	
17	Drake	3TVXtAsR1Inumwj472S9r4	3SpBlxme9WbeQdI9kx7KAV	Certified Lover Boy	
18	Drake	3TVXtAsR1Inumwj472S9r4	6OQ9gBfg5EXeNAEwGSs6jK	Dark Lane Demo Tapes	
19	Drake	3TVXtAsR1Inumwj472S9r4	6OQ9gBfg5EXeNAEwGSs6jK	Dark Lane Demo Tapes	
20	Drake	3TVXtAsR1Inumwj472S9r4	0bNfrGST1xSPZ3V3g5jjHT	What Did I Miss?	
21	Drake	3TVXtAsR1Inumwj472S9r4	6X1x82kppWZmDzlXXK3y3q	Take Care (Deluxe)	
22	Drake	3TVXtAsR1Inumwj472S9r4	6X1x82kppWZmDzlXXK3y3q	Take Care (Deluxe)	
23	Drake	3TVXtAsR1Inumwj472S9r4	6X1x82kppWZmDzlXXK3y3q	Take Care (Deluxe)	
24	Drake	3TVXtAsR1Inumwj472S9r4	6X1x82kppWZmDzlXXK3y3q	Take Care (Deluxe)	
25	Drake	3TVXtAsR1Inumwj472S9r4	4czdORdCWP9umpbhFXK2fW	For All The Dogs	
26	Drake	3TVXtAsR1Inumwj472S9r4	5LuoozUhs2pl3glZeAJl89	Scary Hours 2	
27	Drake	3TVXtAsR1Inumwj472S9r4	40GMAhriYJRO1rsY4YdrZb	Views	
28	Drake	3TVXtAsR1Inumwj472S9r4	1lXY618HWkwYKJWBRYR4MK	More Life	

	artist	artist_id	album_id	album_name	a
29	Drake	3TVXtAsR1Inumwj472S9r4	0v4UlHEDcUBbV9kM3TX4bj	Live For	
30	Drake	3TVXtAsR1Inumwj472S9r4	4VVBDjcpEc8vFWMlQfnmHX	Miss Me	
31	Drake	3TVXtAsR1Inumwj472S9r4	4OQEGCbts7MiJm8wfwC9BX	LOYAL (feat. Drake)	
32	Drake	3TVXtAsR1Inumwj472S9r4	7iLdPmU6eUTwKsYifBQ6zb	TWIST & TURN (feat. Drake & PARTYNEXTDOOR)	
33	Drake	3TVXtAsR1Inumwj472S9r4	4v2wkI8xYxp0gTtSMwbRoj	She Will	
34	Drake	3TVXtAsR1Inumwj472S9r4	4P9NQ88JRt2r39QlsevM1n	Make Me Proud	

Now that I have gathered the necessary stats it is time to visualize the data

In [38]:
```
combined = pd.concat([kendrickStats.assign(artist="Kendrick Lamar"), drakeSt
```

In [39]:
```
combined[["artist", "track_name", "rhyme_density", "word_count", "complexity
```

Out[39]:

	artist	track_name	rhyme_density	word_count	complexity	popularity
0	Kendrick Lamar	Not Like Us	0.897577	907	814.102423	89
1	Kendrick Lamar	HUMBLE.	0.892562	598	533.752066	86
2	Kendrick Lamar	tv off (feat. lefty gunplay)	0.929504	770	715.718016	86
3	Kendrick Lamar	LOVE. FEAT. ZACARI.	0.947242	488	462.254197	85
4	Kendrick Lamar	Swimming Pools (Drank)	0.928654	859	797.713464	84
5	Kendrick Lamar	PRIDE.	0.882353	504	444.705882	82
6	Kendrick Lamar	DNA.	0.875367	671	587.370968	81
7	Kendrick Lamar	Money Trees	0.947232	1153	1092.158304	81
8	Kendrick Lamar	squabble up	0.872340	536	467.574468	81
9	Kendrick Lamar	Alright	0.929379	705	655.211864	81
10	Kendrick Lamar	LOYALTY. FEAT. RIHANNA.	0.954380	591	564.038321	78
11	Kendrick Lamar	Count Me Out	0.925966	956	885.223176	77
12	Kendrick Lamar	peekaboo (feat. azchike)	0.926357	602	557.666667	77
13	Kendrick Lamar	United In Grief	0.885978	646	572.342065	76
14	Kendrick Lamar	euphoria	0.924138	1441	1331.682759	76
15	Kendrick Lamar	King Kunta	0.928471	714	662.928471	75
16	Kendrick Lamar	Poetic Justice	0.942623	981	924.713115	75
17	Kendrick Lamar	Push Thru	0.873418	673	587.810127	16
18	Kendrick Lamar	Malcolm X	0.910781	534	486.356877	27
19	Kendrick Lamar	Bitch, Don’t Kill My Vibe - Remix	0.896181	842	754.584726	28

	artist	track_name	rhyme_density	word_count	complexity	popularity
20	Kendrick Lamar	Two Presidents	0.889465	575	511.442142	34
21	Kendrick Lamar	BLOOD.	0.741379	172	127.517241	38
0	Drake	One Dance	0.934066	469	438.076923	92
1	Drake	Headlines	0.917969	509	467.246094	89
2	Drake	Passionfruit	0.861472	228	196.415584	89
3	Drake	God's Plan	0.864023	349	301.543909	87
4	Drake	Not You Too (feat. Chris Brown)	0.878553	388	340.878553	86
5	Drake	NOKIA	0.943700	784	739.860590	85
6	Drake	Hotline Bling	0.931663	431	401.546697	85
7	Drake	DIE TRYING	0.895377	411	368.000000	84
8	Drake	Teenage Fever	0.862500	319	275.137500	83
9	Drake	9	0.896078	518	464.168627	83
10	Drake	Hold On, We're Going Home	0.996923	339	337.956923	82
11	Drake	Jimmy Cooks (feat. 21 Savage)	0.943704	750	707.777778	82
12	Drake	Nice For What	0.932091	751	700.000000	81
13	Drake	SOMEBODY LOVES ME	0.916442	371	340.000000	81
14	Drake	Which One (feat. Central Cee)	0.937238	566	530.476987	81
15	Drake	Best I Ever Had	0.940334	837	787.059666	81
16	Drake	Trust Issues	0.909256	543	493.725953	81
17	Drake	Yebba’s Heartbreak	0.803150	127	102.000000	81
18	Drake	Time Flies	0.913146	383	349.734742	80
19	Drake	Chicago Freestyle (feat. Giveon)	0.953347	580	552.941176	80
20	Drake	What Did I Miss?	0.892704	466	416.000000	80
21	Drake	Take Care	0.902490	486	438.609959	80
22	Drake	Marvins Room	0.932432	596	555.729730	80
23	Drake	Practice	0.954783	579	552.819130	80

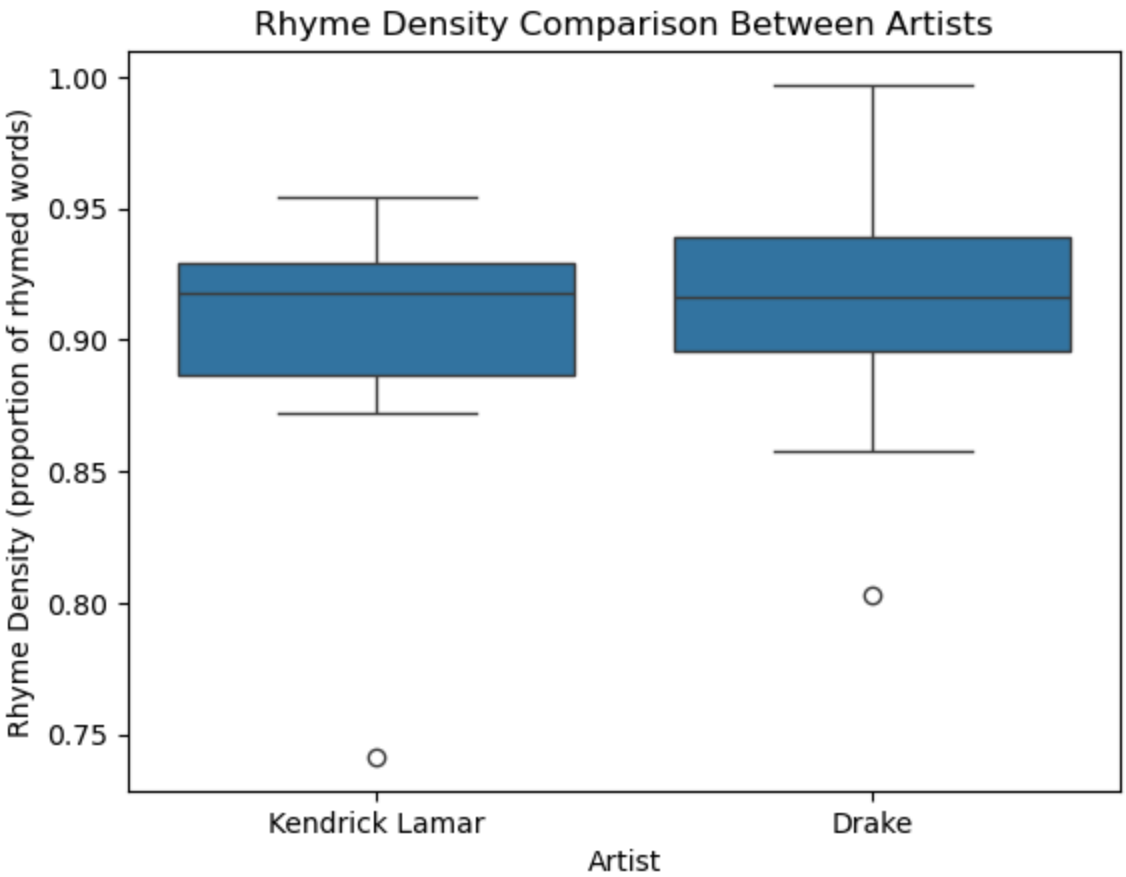
	artist	track_name	rhyme_density	word_count	complexity	popularity
24	Drake	Over My Dead Body	0.911475	608	554.177049	80
25	Drake	Rich Baby Daddy (feat. Sexyy Red & SZA)	0.960729	811	779.151473	80
26	Drake	Wants and Needs (feat. Lil Baby)	0.857977	514	441.000000	79
27	Drake	Feel No Ways	0.955882	340	325.000000	79
28	Drake	Do Not Disturb	0.882622	656	579.000000	79
29	Drake	Live For	0.899115	567	509.798230	29
30	Drake	Miss Me	0.904707	857	775.334099	34
31	Drake	LOYAL (feat. Drake)	0.991228	342	339.000000	37
32	Drake	TWIST & TURN (feat. Drake & PARTYNEXTDOOR)	0.934609	620	579.457735	38
33	Drake	She Will	0.929697	824	766.070303	39
34	Drake	Make Me Proud	0.909091	685	622.727273	42

In [40]:
```
combined.describe()[["rhyme_density", "word_count", "complexity", "popularit
```

Out[40]:

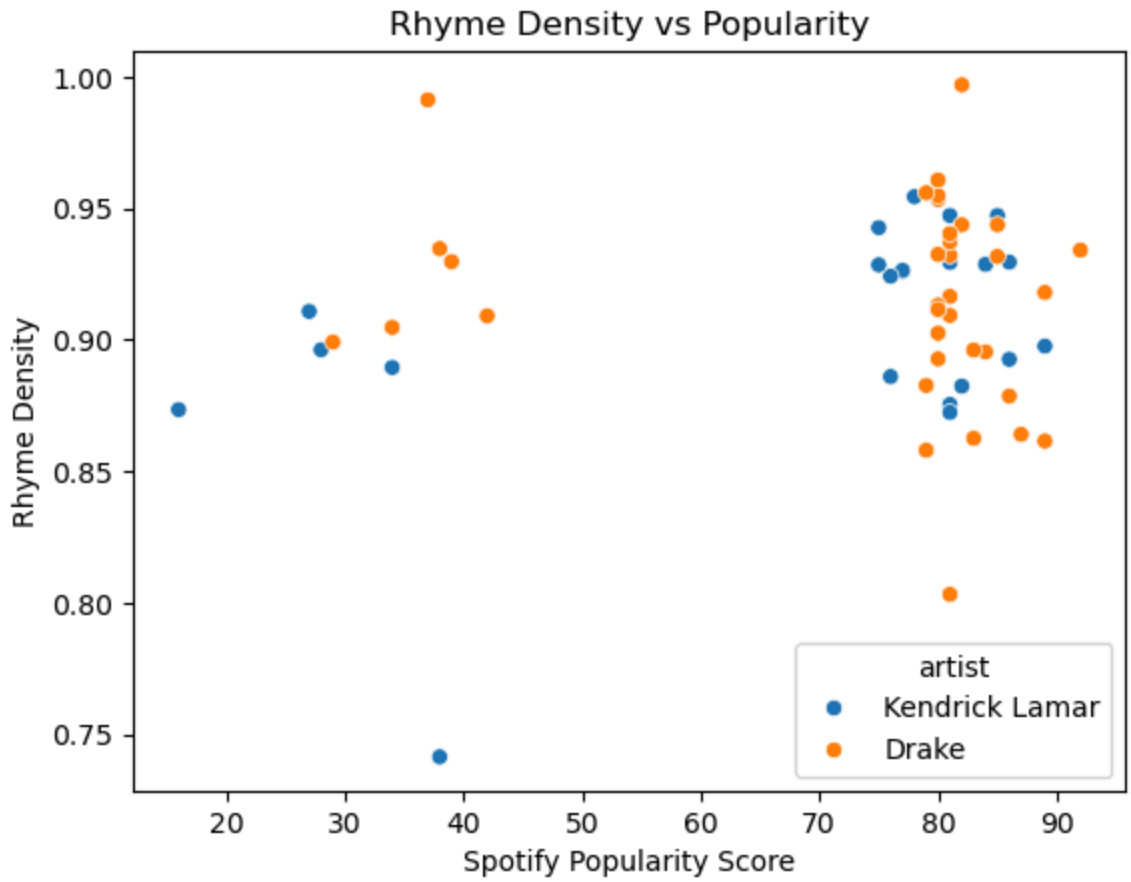
	rhyme_density	word_count	complexity	popularity
count	57.000000	57.000000	57.000000	57.000000
mean	0.911437	605.649123	555.531404	72.315789
std	0.041395	234.576251	221.631553	19.989424
min	0.741379	127.000000	102.000000	16.000000
25%	0.892562	469.000000	438.076923	76.000000
50%	0.916442	580.000000	552.819130	80.000000
75%	0.934609	750.000000	700.000000	83.000000
max	0.996923	1441.000000	1331.682759	92.000000

In [41]:
```
sb.boxplot(x="artist", y="rhyme_density", data=combined)
plt.title("Rhyme Density Comparison Between Artists")
plt.ylabel("Rhyme Density (proportion of rhymed words)")
plt.xlabel("Artist")
plt.show()
```



Both artists show very high rhyme density (Lots 0.9+). Drake's median looks slightly higher, but both distributions overlap heavily. This shows rhyme use is consistently strong in both catalogs.

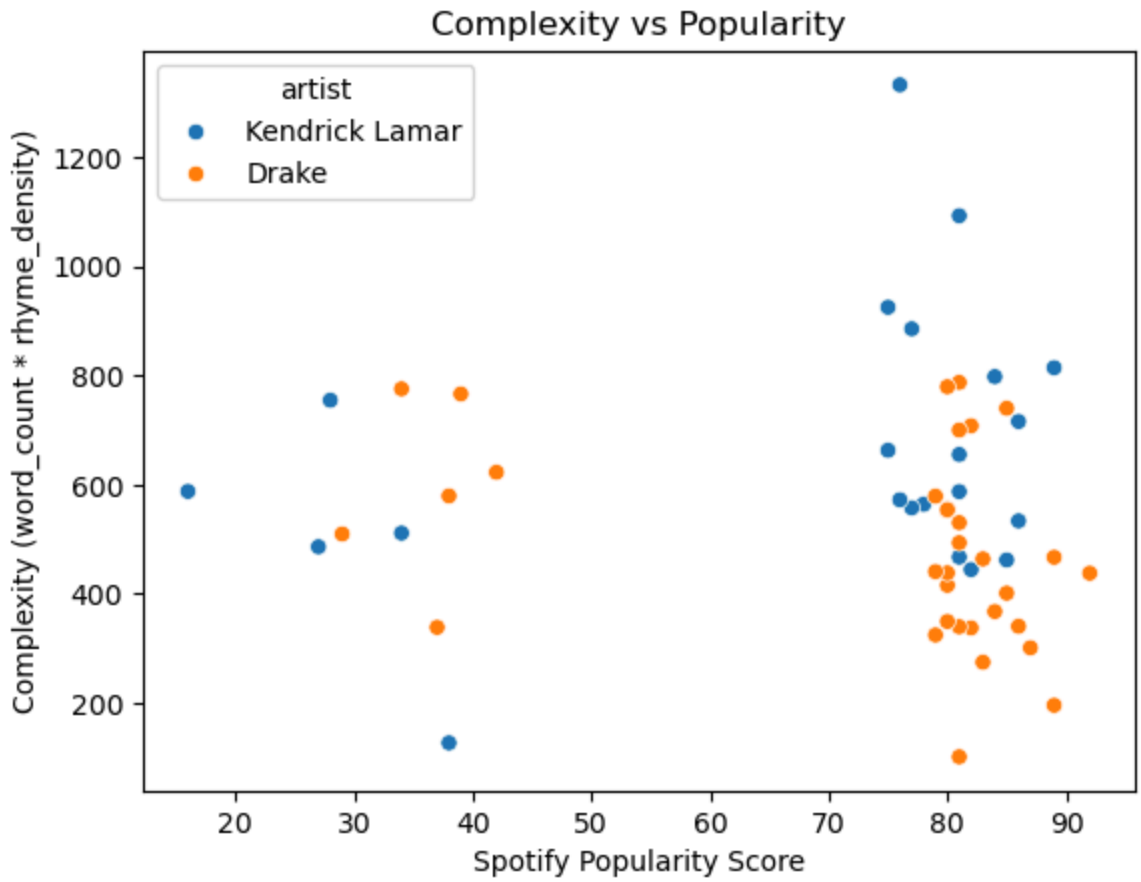
In [42]:
```
sb.scatterplot(x="popularity", y="rhyme_density", hue="artist", data=combine
plt.title("Rhyme Density vs Popularity")
plt.xlabel("Spotify Popularity Score")
plt.ylabel("Rhyme Density")
plt.show()
```



Kendrick: +0.44 -> moderate positive correlation. This suggests his more lyrically complex tracks tend to be more popular. This aligns with his reputation for producing technically dense songs that still resonate well with his audience.

Drake: –0.29 -> weak negative correlation. This shows more rhyming or complex writing slightly correlates with lower popularity. Makes sense given Drake's pop crossover appeal his simpler, melodic songs are usually the hits.

In [43]:
```
sb.scatterplot(x="popularity", y="complexity", hue="artist", data=combined)
plt.title("Complexity vs Popularity")
plt.xlabel("Spotify Popularity Score")
plt.ylabel("Complexity (word_count * rhyme_density)")
plt.show()
```



Correlation magnitudes are smaller (Kendrick +0.25, Drake −0.24). This implies rhyme density matters a bit more than total wordiness.

Now lets look at the actual correlation numbers to double check findings

In [44]:
```
print("Kendrick: rhyme density vs popularity =",
      kendrickStats["rhyme_density"].corr(kendrickStats["popularity"]))

print("Drake: rhyme density vs popularity =",
      drakeStats["rhyme_density"].corr(drakeStats["popularity"]))

print("Kendrick: complexity vs popularity =",
      kendrickStats["complexity"].corr(kendrickStats["popularity"]))

print("Drake: complexity vs popularity =",
      drakeStats["complexity"].corr(drakeStats["popularity"]))
```

```
Kendrick: rhyme density vs popularity = 0.43626748634343915
Drake: rhyme density vs popularity = -0.15539574770524997
Kendrick: complexity vs popularity = 0.2870718851108495
Drake: complexity vs popularity = -0.31819599194826137
```

The data above suggests that Kendrick Lamar's songs show a moderate positive relationship between lyrical density and popularity. This implies that his more lyrically

intricate song-writing tends to perform well with the public. Drake, on the other hand, displays a mild negative correlation between rhyme complexity and popularity. This consistent with his broader pop appeal. Simpler and more melodic tracks dominate his top-performing songs. While both artists exhibit high overall rhyme density, the role of lyrical complexity in driving popularity appears more significant for Kendrick than for Drake.

Thank you for viewing!

Little Bonus Analysis

Since my question has a focus on each artists careers as a whole, I think it is fitting to view these stats across each year of their careers

In [45]:
```
combined["release_year"] = combined["release_date"].str[:4].astype(float) #
combined
```

Out[45]:

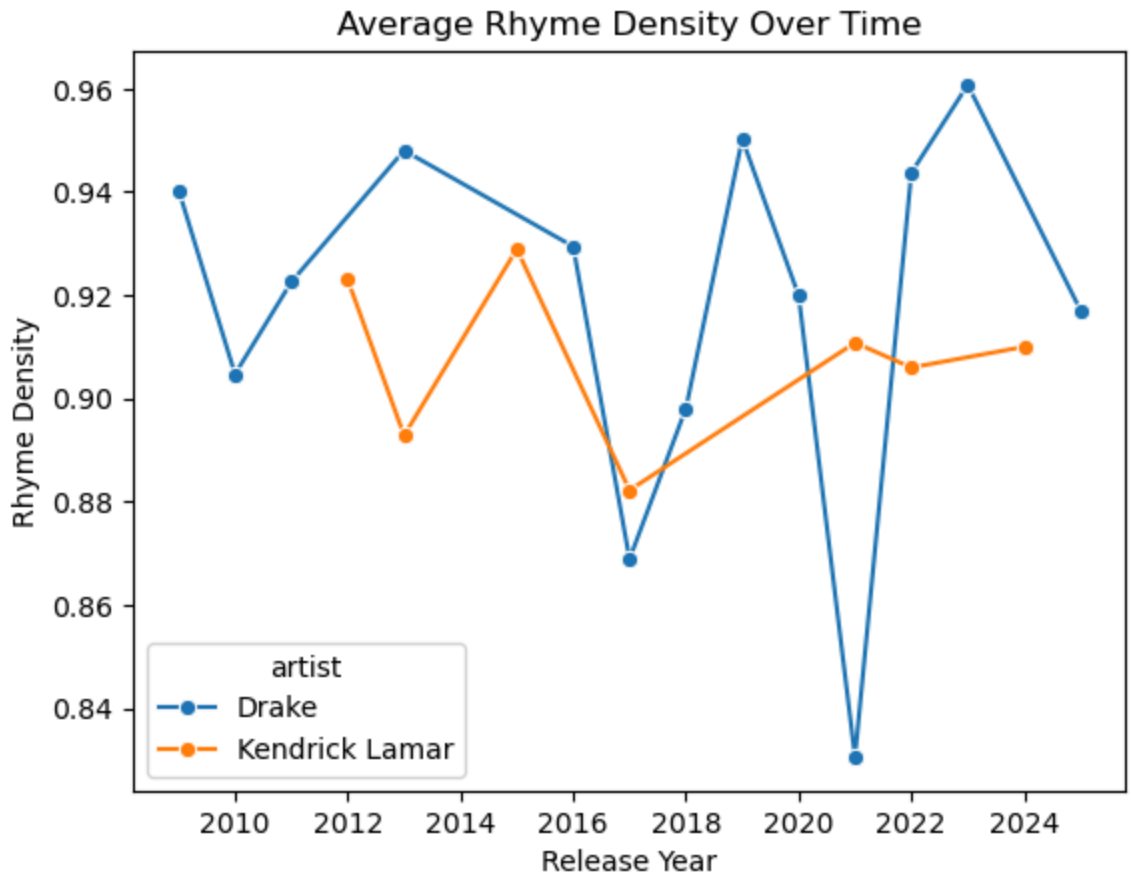
	artist	artist_id	album_id	album_nan
0	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	5JjnoGJyOxfSZUZtk2rRwZ	Not Like U
1	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	4eLPsYPBmXABThSJ821sqY	DAM
2	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	0hvT3yIEysuuvkK73vgdcW	GN
3	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	4eLPsYPBmXABThSJ821sqY	DAM
4	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	2XPrwlaAHHXnJzP9tBcIzH	Swimming Poo (Dran
5	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	4eLPsYPBmXABThSJ821sqY	DAM
6	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	4eLPsYPBmXABThSJ821sqY	DAM
7	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	6PBZN8cbwkqm1ERj2BGXJ1	good kid, m.A.A c
8	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	0hvT3yIEysuuvkK73vgdcW	GN
9	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	7ycBtnsMtyVbbwTfJwRjSP	To Pimp A Butter
10	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	4eLPsYPBmXABThSJ821sqY	DAM
11	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	79ONNoS4M9tfIA1mYLBYVX	Mr. Morale & T Big Steppe
12	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	0hvT3yIEysuuvkK73vgdcW	GN
13	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	79ONNoS4M9tfIA1mYLBYVX	Mr. Morale & T Big Steppe
14	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	32bR4LcEc1PvJEhaKoo4ZN	euphor

	artist	artist_id	album_id	album_nan
15	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	7ycBtnsMtyVbbwTfJwRjSP	To Pimp A Butter
16	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	3DGQ1iZ9XKUQxAUWjfC34w	good kid, m.A.A city (Delux
17	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	3qHsYfN7ua1WGhQn1wquIn	Push Thru (fea Kendrick Lam and Curren$
18	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	4NuKTiNjwcX4ZG7TXEBnYh	Malcolm
19	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	4XF4wNh0WHyrayqCYS8xY0	Bitch, Don't Kill M Vibe (Rem
20	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	5moHeRF6YwQz9Mhc1SB0v0	Two Presider
21	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	4alcGHjstaALJHHiljfy3H	DAM COLLECTOR EDITIO
0	Drake	3TVXtAsR1Inumwj472S9r4	40GMAhriYJRO1rsY4YdrZb	View
1	Drake	3TVXtAsR1Inumwj472S9r4	6X1x82kppWZmDzlXXK3y3q	Take Care (Delux
2	Drake	3TVXtAsR1Inumwj472S9r4	1lXY618HWkwYKJWBRYR4MK	More L
3	Drake	3TVXtAsR1Inumwj472S9r4	1ATL5GLyefJaxhQzSPVrLX	Scorpi
4	Drake	3TVXtAsR1Inumwj472S9r4	6OQ9gBfg5EXeNAEwGSs6jK	Dark Lane Der Tap
5	Drake	3TVXtAsR1Inumwj472S9r4	6Rl6YoCarF2GHPSQmmFjuR	*ome*xy $ongs 4
6	Drake	3TVXtAsR1Inumwj472S9r4	40GMAhriYJRO1rsY4YdrZb	View
7	Drake	3TVXtAsR1Inumwj472S9r4	6Rl6YoCarF2GHPSQmmFjuR	*ome*xy $ongs 4

	artist	artist_id	album_id	album_nan
8	Drake	3TVXtAsR1Inumwj472S9r4	1lXY618HWkwYKJWBRYR4MK	More L
9	Drake	3TVXtAsR1Inumwj472S9r4	40GMAhriYJRO1rsY4YdrZb	Vie
10	Drake	3TVXtAsR1Inumwj472S9r4	2ZUFSbIkmFkGag000RWOpA	Nothing Was T Same (Delux
11	Drake	3TVXtAsR1Inumwj472S9r4	3cf4iSSKd8ffTncbtKljXw	Honest Nevermi
12	Drake	3TVXtAsR1Inumwj472S9r4	1ATL5GLyefJaxhQzSPVrLX	Scorpi
13	Drake	3TVXtAsR1Inumwj472S9r4	6Rl6YoCarF2GHPSQmmFjuR	*ome*exy $ongs 4
14	Drake	3TVXtAsR1Inumwj472S9r4	4TWJJZmgYm6tE6PzqpluQ0	Which O
15	Drake	3TVXtAsR1Inumwj472S9r4	1LShhEEKRT5MNPcO7jtYHh	So Far Go
16	Drake	3TVXtAsR1Inumwj472S9r4	7dqpveMVcWgbzqYrOdkFTD	Care Packa
17	Drake	3TVXtAsR1Inumwj472S9r4	3SpBlxme9WbeQdI9kx7KAV	Certified Lover B
18	Drake	3TVXtAsR1Inumwj472S9r4	6OQ9gBfg5EXeNAEwGSs6jK	Dark Lane Der Tap
19	Drake	3TVXtAsR1Inumwj472S9r4	6OQ9gBfg5EXeNAEwGSs6jK	Dark Lane Der Tap
20	Drake	3TVXtAsR1Inumwj472S9r4	0bNfrGST1xSPZ3V3g5jjHT	What Did I Mis
21	Drake	3TVXtAsR1Inumwj472S9r4	6X1x82kppWZmDzlXXK3y3q	Take Care (Delux
22	Drake	3TVXtAsR1Inumwj472S9r4	6X1x82kppWZmDzlXXK3y3q	Take Care (Delux

	artist	artist_id	album_id	album_nan
23	Drake	3TVXtAsR1Inumwj472S9r4	6X1x82kppWZmDzlXXK3y3q	Take Care (Delux
24	Drake	3TVXtAsR1Inumwj472S9r4	6X1x82kppWZmDzlXXK3y3q	Take Care (Delux
25	Drake	3TVXtAsR1Inumwj472S9r4	4czdORdCWP9umpbhFXK2fW	For All The Do
26	Drake	3TVXtAsR1Inumwj472S9r4	5LuoozUhs2pl3glZeAJl89	Scary Hours
27	Drake	3TVXtAsR1Inumwj472S9r4	40GMAhriYJRO1rsY4YdrZb	Viev
28	Drake	3TVXtAsR1Inumwj472S9r4	1lXY618HWkwYKJWBRYR4MK	More L
29	Drake	3TVXtAsR1Inumwj472S9r4	0v4UlHEDcUBbV9kM3TX4bj	Live F
30	Drake	3TVXtAsR1Inumwj472S9r4	4VVBDjcpEc8vFWMlQfnmHX	Miss M
31	Drake	3TVXtAsR1Inumwj472S9r4	4OQEGCbts7MiJm8wfwC9BX	LOYAL (fea Drak
32	Drake	3TVXtAsR1Inumwj472S9r4	7iLdPmU6eUTwKsYifBQ6zb	TWIST & TUF (feat. Drake PARTYNEXTDOO
33	Drake	3TVXtAsR1Inumwj472S9r4	4v2wkI8xYxp0gTtSMwbRoj	She W
34	Drake	3TVXtAsR1Inumwj472S9r4	4P9NQ88JRt2r39QlsevM1n	Make Me Prou

In [46]:
```
yearly = (combined.groupby(["artist", "release_year"])["rhyme_density"].mear
sb.lineplot(data=yearly, x="release_year", y="rhyme_density", hue="artist",
plt.title("Average Rhyme Density Over Time")
plt.xlabel("Release Year")
plt.ylabel("Rhyme Density")
plt.show()
```



In [47]:
```
combined
```

Out[47]:

	artist	artist_id	album_id	album_nan
0	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	5JjnoGJyOxfSZUZtk2rRwZ	Not Like U
1	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	4eLPsYPBmXABThSJ821sqY	DAM
2	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	0hvT3yIEysuuvkK73vgdcW	GN
3	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	4eLPsYPBmXABThSJ821sqY	DAM
4	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	2XPrwlaAHHXnJzP9tBcIzH	Swimming Poo (Dran
5	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	4eLPsYPBmXABThSJ821sqY	DAM
6	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	4eLPsYPBmXABThSJ821sqY	DAM
7	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	6PBZN8cbwkqm1ERj2BGXJ1	good kid, m.A.A c
8	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	0hvT3yIEysuuvkK73vgdcW	GN
9	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	7ycBtnsMtyVbbwTfJwRjSP	To Pimp A Butter
10	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	4eLPsYPBmXABThSJ821sqY	DAM
11	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	79ONNoS4M9tfIA1mYLBYVX	Mr. Morale & T Big Steppe
12	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	0hvT3yIEysuuvkK73vgdcW	GN
13	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	79ONNoS4M9tfIA1mYLBYVX	Mr. Morale & T Big Steppe
14	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	32bR4LcEc1PvJEhaKoo4ZN	euphor

	artist	artist_id	album_id	album_nan
15	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	7ycBtnsMtyVbbwTfJwRjSP	To Pimp A Butter
16	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	3DGQ1iZ9XKUQxAUWjfC34w	good kid, m.A.A city (Delux
17	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	3qHsYfN7ua1WGhQn1wquIn	Push Thru (fea Kendrick Lam and Curren$
18	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	4NuKTiNjwcX4ZG7TXEBnYh	Malcolm
19	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	4XF4wNh0WHyrayqCYS8xY0	Bitch, Don't Kill M Vibe (Rem
20	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	5moHeRF6YwQz9Mhc1SB0v0	Two Presider
21	Kendrick Lamar	2YZyLoL8N0Wb9xBt1NhZWg	4alcGHjstaALJHHiljfy3H	DAM COLLECTOR EDITIO
0	Drake	3TVXtAsR1Inumwj472S9r4	40GMAhriYJRO1rsY4YdrZb	View
1	Drake	3TVXtAsR1Inumwj472S9r4	6X1x82kppWZmDzlXXK3y3q	Take Care (Delux
2	Drake	3TVXtAsR1Inumwj472S9r4	1lXY618HWkwYKJWBRYR4MK	More L
3	Drake	3TVXtAsR1Inumwj472S9r4	1ATL5GLyefJaxhQzSPVrLX	Scorpi
4	Drake	3TVXtAsR1Inumwj472S9r4	6OQ9gBfg5EXeNAEwGSs6jK	Dark Lane Der Tap
5	Drake	3TVXtAsR1Inumwj472S9r4	6Rl6YoCarF2GHPSQmmFjuR	*ome*xy $ongs 4
6	Drake	3TVXtAsR1Inumwj472S9r4	40GMAhriYJRO1rsY4YdrZb	View
7	Drake	3TVXtAsR1Inumwj472S9r4	6Rl6YoCarF2GHPSQmmFjuR	*ome*xy $ongs 4

	artist	artist_id	album_id	album_nan
8	Drake	3TVXtAsR1Inumwj472S9r4	1lXY618HWkwYKJWBRYR4MK	More L
9	Drake	3TVXtAsR1Inumwj472S9r4	40GMAhriYJRO1rsY4YdrZb	Vie
10	Drake	3TVXtAsR1Inumwj472S9r4	2ZUFSbIkmFkGag000RWOpA	Nothing Was T Same (Delux
11	Drake	3TVXtAsR1Inumwj472S9r4	3cf4iSSKd8ffTncbtKljXw	Honest Nevermi
12	Drake	3TVXtAsR1Inumwj472S9r4	1ATL5GLyefJaxhQzSPVrLX	Scorpi
13	Drake	3TVXtAsR1Inumwj472S9r4	6Rl6YoCarF2GHPSQmmFjuR	*ome*exy $ongs 4
14	Drake	3TVXtAsR1Inumwj472S9r4	4TWJJZmgYm6tE6PzqpluQ0	Which O
15	Drake	3TVXtAsR1Inumwj472S9r4	1LShhEEKRT5MNPcO7jtYHh	So Far Go
16	Drake	3TVXtAsR1Inumwj472S9r4	7dqpveMVcWgbzqYrOdkFTD	Care Packa
17	Drake	3TVXtAsR1Inumwj472S9r4	3SpBlxme9WbeQdI9kx7KAV	Certified Lover B
18	Drake	3TVXtAsR1Inumwj472S9r4	6OQ9gBfg5EXeNAEwGSs6jK	Dark Lane Der Tap
19	Drake	3TVXtAsR1Inumwj472S9r4	6OQ9gBfg5EXeNAEwGSs6jK	Dark Lane Der Tap
20	Drake	3TVXtAsR1Inumwj472S9r4	0bNfrGST1xSPZ3V3g5jjHT	What Did I Mis
21	Drake	3TVXtAsR1Inumwj472S9r4	6X1x82kppWZmDzlXXK3y3q	Take Care (Delux
22	Drake	3TVXtAsR1Inumwj472S9r4	6X1x82kppWZmDzlXXK3y3q	Take Care (Delux

	artist	artist_id	album_id	album_nan
23	Drake	3TVXtAsR1Inumwj472S9r4	6X1x82kppWZmDzlXXK3y3q	Take Care (Delux
24	Drake	3TVXtAsR1Inumwj472S9r4	6X1x82kppWZmDzlXXK3y3q	Take Care (Delux
25	Drake	3TVXtAsR1Inumwj472S9r4	4czdORdCWP9umpbhFXK2fW	For All The Do
26	Drake	3TVXtAsR1Inumwj472S9r4	5LuoozUhs2pl3glZeAJl89	Scary Hours
27	Drake	3TVXtAsR1Inumwj472S9r4	40GMAhriYJRO1rsY4YdrZb	Viev
28	Drake	3TVXtAsR1Inumwj472S9r4	1lXY618HWkwYKJWBRYR4MK	More L
29	Drake	3TVXtAsR1Inumwj472S9r4	0v4UlHEDcUBbV9kM3TX4bj	Live F
30	Drake	3TVXtAsR1Inumwj472S9r4	4VVBDjcpEc8vFWMlQfnmHX	Miss M
31	Drake	3TVXtAsR1Inumwj472S9r4	4OQEGCbts7MiJm8wfwC9BX	LOYAL (fea Drak
32	Drake	3TVXtAsR1Inumwj472S9r4	7iLdPmU6eUTwKsYifBQ6zb	TWIST & TUF (feat. Drake PARTYNEXTDOO
33	Drake	3TVXtAsR1Inumwj472S9r4	4v2wkI8xYxp0gTtSMwbRoj	She W
34	Drake	3TVXtAsR1Inumwj472S9r4	4P9NQ88JRt2r39QlsevM1n	Make Me Prou

Both artists maintain pretty high levels of lyrical complexity throughout their careers. Drake's rhyme density fluctuates more, suggesting variation across releases, while Kendrick Lamar's appears more stable and slightly trends upward, reflecting more lyrical refinement over time.

In [48]:
```
combined['release_year'] = combined['release_year'].astype(int)
```

In [49]:
```
project2ReleaseYears = combined[['track_name', 'artist', 'release_year']]
```

In [50]:
```
project2ReleaseYears.to_csv('project2ReleaseYears.csv', index=False)
```